Anna Rahn (anna.rahn@colorado.edu)
Broderick Schmidt (broderick.schmidt@colorado.edu)
Kieran Stone (kieran.stone@colorado.edu)

Clustering fNIRS Neuroimaging Data for Task Classification

Majid Zamani's CSCI 4022

Abstract

Functional near-infrared spectroscopy (fNIRS) is a non-invasive, portable neuroimaging technique to measure the changes in oxygenation concentration on the cortical surface of the brain. We used an open-access dataset where participants tapped fingers on their right hand, tapped fingers on their left hand, or tapped their foot, to explore the applications of k-means clustering to task classification. We found that using higher dimensionality improved the accuracy of prediction when clustered over an aggregated dataset of all participants, that aggregating all participants decreases the variability, and that foot-tapping consistently had the highest prediction accuracy compared to left-hand finger-tapping and right-hand finger-tapping. We also found that linear dimension reduction from 120 features to 75 features did not significantly improve the accuracy of classification, but that using a neural network classifier (a multi-layer perceptron) consistently produced better results. Suggested future directions include optimizing the feature extraction/selection process, applying clustering to finding significant channels, and applying these methods to classify data in other regions of the brain.

Introduction

Background

Functional near-infrared spectroscopy (fNIRS) is a portable, non-invasive method of measuring the brain's changes in hemoglobin concentrations using infrared light. Per a journal entry from the National Library of Medicine, the technology "is safe and portable, suiting laboratory and ecologically valid contexts while offering resistance to motion artifacts. Consequently, fNIRS has provided opportunities to work with diverse populations and settings in ways alternative neuroimaging modalities are limited or incompatible." [VII]

Real-world Impact of fNIRS Research

Applications in health care include its use to diagnose and monitor cognitive disorders such as ADHD, Alzheimer's disease, and schizophrenia. [XI] [XII] [XII] By analyzing fNIRS scans in clinical patients, we can better understand disease progression and evaluate more suitable treatment responses.

fNIRS has also found itself a role in assisting individuals with disabilities through the use of fNIRS-based Brain-Computer Interfaces (BCIs). Studies have shown that hybrid

EEG-fNIRS-based BCIs can be used to control devices that can replicate communication and movement. [V] Applying advanced data science clustering algorithms to the data from these instruments can help to more accurately decipher the results. [VI]

Project Description

This study will apply the k-means clustering algorithm to an open-access fNIRS dataset to find which trials are most similar to one another. We expect the resulting clusters to predict whether the event in the observed trial was left-hand finger-tapping, right-hand finger-tapping, or foot-tapping. Clustering has been used to classify incoming data by task [XIV][XV], which is the application we will be exploring, and to classify channels by activation [XVI].

We are primarily interested in this dataset for the applications to our team member's current research with the Brain, AI, and Child Lab, which lies in the intersection of artificial intelligence (AI) and machine learning (ML), human-computer interaction, and neuroimaging. The lab is currently interested in classifying incoming data by task [X], classifying channels by activation, and generating synthetic data; this project has allowed us to explore models relevant to this research using a larger dataset.

Data

The open-access fNIRS dataset accessible on Github [VIII][IX] consists of 30 voluntary participants. Each participant's cerebral hemoglobin patterns were tracked and quantified over the course of 75 trials. Of those 75 trials, 25 involved the participant taking part in left-hand unilateral complex tapping, 25 of them involved the participant taking part in right-hand unilateral complex tapping, and 25 of them involved the participant tapping their foot. Each of these trials involves the collection of data through 20 channels: 10 channels on the left cerebral hemisphere and 10 on the right. Each channel reports concentration changes for both oxygenated and deoxygenated hemoglobin. After preprocessing the data (using bandpass filtering, segmentation, and baseline correction), the authors extract features that consist of three epochs: spans of time from 0 - 5 seconds, 5 - 10 seconds, and 10 - 15 seconds of the trial. The block average of each of these epochs are saved for each trial and channel, producing a [75 x 120] dimension feature matrix for each participant.

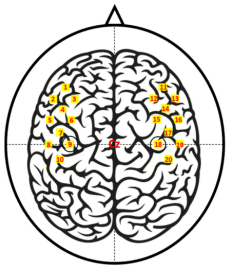

Figure 1. Diagram of channels

The original MATLAB data can be found in our repository in the Data/MATLAB folder. The feature vectors and ground truth produced by the classification script provided by the authors can be found at Data/Clustering - see tutorial_classification.m for details regarding the generation. Finally, the feature vectors for individual participants as rearranged into [75, 40, 3] numpy arrays can be found in Data/NP. When aggregated over all participants, we produce a [2250, 40, 3] numpy array (as each of our thirty participants contains 75 trials).

Methods

Clustering

Our first method of analysis was the k-means algorithm, an unsupervised ML method used for partitional clustering. The steps are as follows:

1. Pick k random centroids
2. Assign each point in the dataset to the closest centroid using Euclidean distance
3. Compute the new centroid of each cluster
4. Repeat steps 1-3 until convergence is reached
5. Calculate the accuracy (as described in Accuracy Clustering) of the clustering
6. Repeat steps 1-6 i number of times, saving the clustering with the highest accuracy

Here, we use $k = 3$ since the number of possible activities is known. We use $i = 25$, finding diminishing returns with greater values.

We apply this clustering algorithm to predict the activity when given three different aggregations of data: (1) a single channel (either oxygenated or deoxygenated) from a single participant, (2) all channels (both oxygenated and deoxygenated) from a single participant, and (3) all channels from all participants.

This hard clustering model is useful because the feature matrix is dense and the distances between trials are easily computed with a Euclidean distance function. Activities that are similar to one another should have similar patterns of activation (e.g. left-hand finger-tapping should produce more oxygenation in the right cortex), so clustering should serve as a proxy for activity classification.

Dimension Reduction

Our original k-means algorithm clusters over 120 dimensions (40 channel dimensions, each with three block averages). In order to investigate whether linear dimension reduction can improve the accuracy of this method, we reduce our number of features to 75 by using principal component analysis (PCA). First, the algorithm standardizes the data by subtracting the mean and dividing it by the standard deviation. Then it computes the covariance matrix for the standardized data and computes the eigenvalues and eigenvectors of the covariance matrix. Finally, the algorithm sorts the eigenvectors by their corresponding eigenvalues in descending order. The principal components are the eigenvectors with the largest eigenvalues.

Accuracy Calculation

For each trial, a ground-truth label is provided in the open-access dataset. The accuracy is calculated as the percentage of trials that are correctly labeled by the clustering algorithm out of the total number of trials.

Multi-Layer Perceptron

In order to compare the accuracy of k-means clustering to deep learning classification, we implement a multi-layer perceptron using Python's scikit-learn package. We use 1000 maximum iterations and the default parameter of 100 hidden layers. Then, we divide the aggregation of the total dataset (all participants, using all channels) into a training set and validation set. After training on 90% of the data, distributed evenly across participants and types of activity, we use the model to predict the task of each trial in the validation set. Accuracy is calculated in the same way as k-means clustering (described in Accuracy Calculation).

Results

We first partitioned the data by channel, then clustered by trial. In other words, we tried to predict whether any given trial was right-hand tapping, left-hand tapping, or foot-tapping when given the block averages for a single channel across all thirty participants. The accuracy ranged from 35% to 50% (see Figure 2), with great variability between runs, but consistently scored higher than random chance. We then ran a similar prediction but with all channel data from a single participant rather than just one. The additional data led to great performance improvements, ranging from 35% to 80% when clustered by individual participant (see Figure 3). We also noted that some participants, such as 15 and 26, scored consistently higher than others (see Figure 3 and Figure 6).

In addition to calculating accuracy of clustering for individual participants, we also aggregated the data over all participants. The accuracy of clustering tended to be lower, but more consistent between runs. This allowed us to calculate total accuracy by task (see Figure 4). We found that foot-tapping was consistently classified better than finger-tapping on either hand.

Finally, we compared the results of k-means clustering to classification by an artificial neural network (see Figure 5). We found that the aggregate accuracy for the multi-layer perceptron consistently outperformed k-means clustering as a method of classification, often producing double the accuracy.

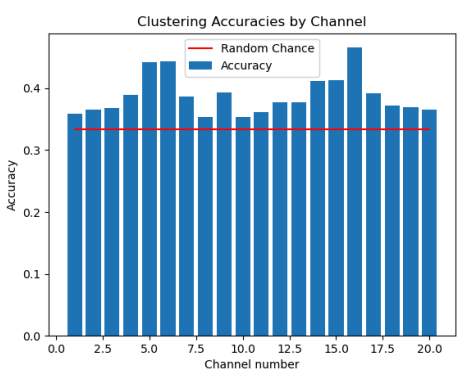

Figure 2. Accuracy of k-means clustering across channels

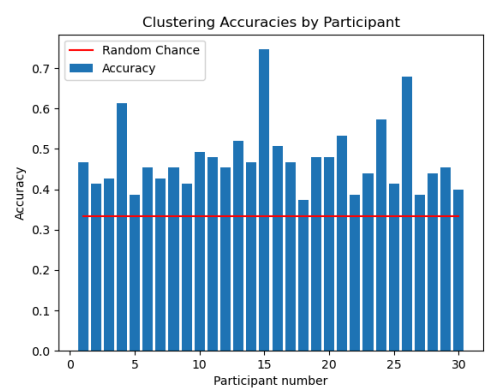

Figure 3. Accuracy of k-means clustering across participants

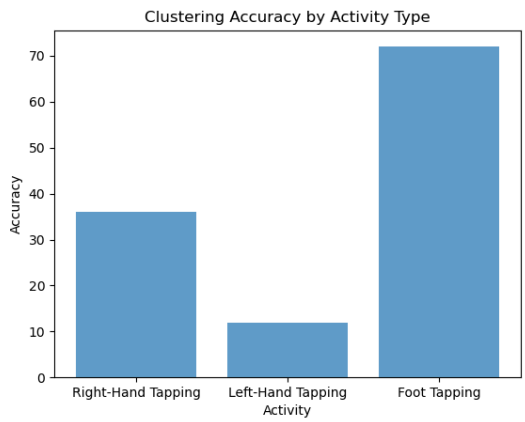

Figure 4. Accuracy of k-means clustering by type of activity

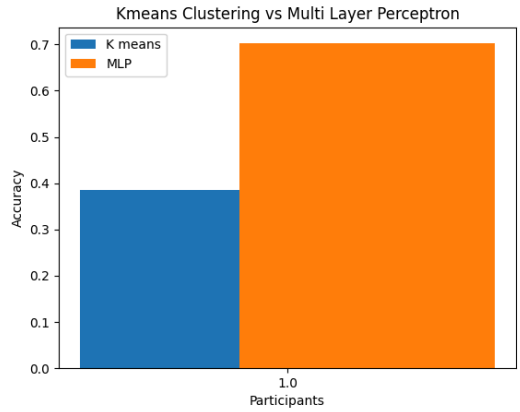

Figure 5. Accuracy of k-means clustering compared to neural network classification

Finally, we compared the accuracy of the reduced dimension data with the non-reduced dimension data and found that reducing the number of dimensions in the data did not significantly improve the clustering accuracy as shown in Figure 6. There was a mixture of improvements and decreases in performance across participants with large variability between runs, so we found the method to be unreliable.

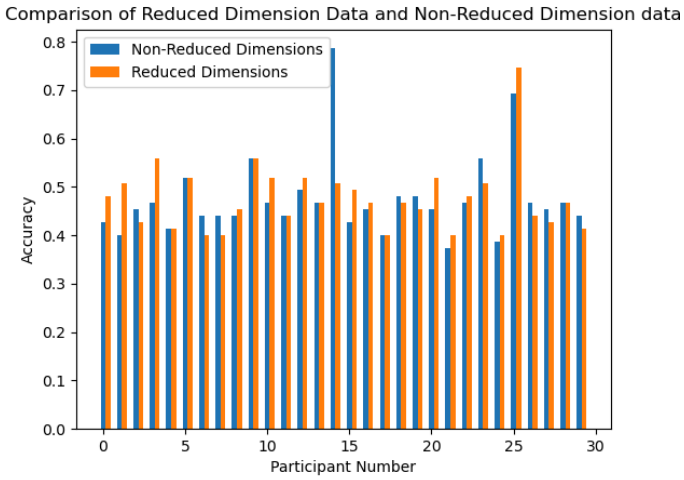

Figure 6. Accuracy of k-means clustering with reduced dimension and non-reduced dimensions

Discussion

When comparing Figures 2 and 3, it is clear that a greater amount of data is positively correlated with a greater clustering accuracy. This may be because the channels are noisy - providing more data can help reduce the overall variance. This finding is also true when comparing the results for a single participant versus the aggregated dataset of all thirty participants; there was significantly more variability in our results between runs for the single participant.

When aggregating all the participants and channels (to produce our [2250 x 40 x 3] feature vector), we found foot-tapping was clustered more accurately than finger-tapping on either hand (see Figure 4). This is in line with our predictions; foot-tapping is more different than finger-tapping and also uses a larger part of the body, and therefore, the activations produced appeared the most distinct. It also makes sense that the accuracy and variance both decreased, as compared to clustering by individual participants; individuals may have distinctly different baseline patterns, making it difficult for the clustering to generalize over such a high dimensionality dataset. However, providing a greater amount of data helps reduce the effects of noise.

We also found that k-means clustering did not outperform deep learning as a method of classification. This is likely due to limitations regarding the shape k-means clustering produces, which assumes circular shapes when using Euclidean distance. Additionally, a neural network may be more suited for working with high-dimensional feature vectors.

Future Work

Improvements to the K-Means Algorithm

Given the size of the feature matrix, we suggest exploring dimensionality reduction and more optimal feature extraction to reduce the computational requirements and noise. Additionally, feature selection (as opposed to extraction) for the dimension reduction may be beneficial because it preserves the exact values of the data, reducing the amount of error produced by the reduction. We also suggest applying clustering to predicting significant channels, improving the baseline correction methodology (as there should be low variability between baseline participant hemoglobin levels after sufficient preprocessing), and creating new data visualizations. Overall, the results seem promising - the clustering accuracy is high for select participants, so we hope that more optimal feature selection and improved preprocessing will continue to increase the accuracy of k-means clustering.

Applications to Other Datasets

Based on the results, we are interested in applying this method of clustering to current research studies on emotional processing in the prefrontal cortex [X]. The lack of hair in this area makes the data collection process more efficient, and we predict the data to be more consistent. We are also interested in comparing the current clustering methods to alternative models, such as support vector machines as used in the paper, as an example of clustering [VIII].

Additionally, we hope to use other clustering methods, such as Gaussian Mixture Models (GMMs), as generative models. BAIC is currently exploring autoencoders as a source of

synthetic data but would find it interesting to compare these two methods for realism and accuracy.

References

I. Bak, S., Park, J., Shin, J., & Jeong, J. (2019). Open-Access FNIRS Dataset for Classification of Unilateral Finger- and Foot-Tapping. Electronics, 8(12), 1486. https://doi.org/10.3390/electronics8121486

II. figshare. (2020, March 7). Open access fNIRS dataset for classification of the unilateral finger- and foot-tapping. Figshare. https://figshare.com/articles/dataset/Open_access_fNIRS_dataset_for_classification_of_the_unilateral_finger-_and_foot-tapping/9783755/1

III. Saikia, M. J. (2023). K-Means Clustering Machine learning approach reveals groups of homogeneous individuals with unique brain activation, task, and performance dynamics using FNIRS. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 31, 2535–2544. https://doi.org/10.1109/tnsre.2023.3278268

IV. Phillips, Z., Canoy, R. J., Paik, S., Lee, S. H., & Kim, B. (2023). Functional Near-Infrared Spectroscopy as a personalized digital healthcare tool for brain monitoring. Journal of Clinical Neurology/˜the œJournal of Clinical Neurology, 19(2), 115. https://doi.org/10.3988/jcn.2022.0406

V. Khan, H., Naseer, N., Yazidi, A., Eide, P. K., Hassan, H. W., & Mirtaheri, P. (2021). Analysis of human GAIT using Hybrid EEG-FNIRS-Based BCI System: A review. Frontiers in Human Neuroscience, 14. https://doi.org/10.3389/fnhum.2020.613254

VI. Naseer, N., & Hong, K. (2015). fNIRS-based brain-computer interfaces: a review. Frontiers in Human Neuroscience, 9. https://doi.org/10.3389/fnhum.2015.00003

VII. Kinder, K. T., Heim, H., Parker, J., Lowery, K., McCraw, A., Eddings, R. N., Defenderfer, J., Sullivan, J. A., & Buss, A. T. (2022). Systematic review of fNIRS studies reveals inconsistent chromophore data reporting practices. Neurophotonics, 9(04). https://doi.org/10.1117/1.nph.9.4.040601

VIII. https://www.researchgate.net/publication/337783882_Open-Access_fNIRS_Dataset_for_Classification_of_Unilateral_Finger-and_Foot-Tapping

A. Open access dataset paper

IX. https://figshare.com/articles/dataset/Open_access_fNIRS_dataset_for_classification_of_the_unilateral_finger-_and_foot-tapping/9783755/1

A. Data location

X. https://dl.acm.org/doi/10.1145/3585088.3593869

A. Empathy conference paper

XI. Yang, J., Ji, X., Quan, W., Liu, Y., Wei, B., & Wu, T. (2020). Classification of schizophrenia by functional connectivity strength using functional near infrared spectroscopy. Frontiers in Neuroinformatics, 14. https://doi.org/10.3389/fninf.2020.00040

XII. Li, R., Rui, G., Chen, W., Li, S., Schulz, P. E., & Zhang, Y. (2018). Early detection of Alzheimer's disease using non-invasive Near-Infrared Spectroscopy. Frontiers in Aging Neuroscience, 10. https://doi.org/10.3389/fnagi.2018.00366

XIII. Yang, C., Shin, J., Kim, J. I., Lim, Y. B., Park, S. H., & Kim, B. (2023). Classifying Children with ADHD Based on Prefrontal Functional Near-infrared Spectroscopy Using Machine Learning. Clinical Psychopharmacology and Neuroscience/Clinical

Psychopharmacology and Neuroscience, 21(4), 693–700.
https://doi.org/10.9758/cpn.22.1025

XIV. https://ui.adsabs.harvard.edu/abs/2021SPIE11629E..2MS/abstract

XV. https://www.researchgate.net/publication/370950875_K-means_Clustering_Machine_Learning_Approach_Reveals_Groups_of_Homogeneous_Individuals_with_Unique_Brain_Activation_Task_and_Performance_Dynamics_using_fNIRS

XVI. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4354588/